

Topic 1.9 - Rational Functions and Vertical Asymptotes

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Identify vertical asymptotes after cancellation and distinguish them from holes.
- Determine one-sided infinite behavior using sign and multiplicity.
- Construct and interpret rational functions with specified asymptotic behavior.

Section A: Foundation and representation

Question 1

Foundation

No calculator

Find holes and vertical asymptotes of $\frac{(x-2)(x+1)}{(x-2)(x-5)}$.

Question 2

Factoring

No calculator

Find vertical asymptotes of $\frac{x^2+1}{(x-3)^2(x+4)}$.

Section A continued

Question 3

One-sided limits

No calculator

Determine one-sided behavior of $\frac{2}{x+1}$ at -1 .

Question 4

Multiplicity

No calculator

Determine one-sided behavior of $\frac{-3}{(x-4)^2}$ at 4 .

Question 5

Sign analysis

No calculator

Analyze $\frac{x+5}{(x-1)^3}$ at 1 .

Section B: Application and reasoning**Question 6**

Sign analysis

No calculator

Analyze $\frac{x-2}{(x+3)^2}$ at -3 .

Question 7

Multiple factors

No calculator

For $\frac{(x-1)(x+2)}{(x+2)^2(x-4)}$, classify $-2, 4$ and analyze one-sided behavior.

Question 8

Construction

No calculator

Construct a function with vertical asymptote 2 , both sides $+\infty$, and zero -1 .

Section C: Challenge and error analysis**Question 9**

Error analysis

No calculator

A student says every denominator zero is a vertical asymptote. Give a counterexample and explain.

Question 10

Parameter

No calculator

Find k so $\frac{k}{(x+2)^2}$ approaches $-\infty$ on both sides of -2 .

Question 11

Context

No calculator

For $D(t) = \frac{500}{t-6} + 80$ with $t > 6$, describe behavior near the domain boundary and as $t \rightarrow \infty$.

Question 12

Challenge

No calculator

Construct a function with VAs at -1 and 2 , with both sides down at -1 and opposite signs left down/right up at 2 .

Section D: AP-style multiple choice

MCQ 1

No calculator

For $\frac{x-1}{(x+2)^2}$ near $x = -2$, both sides approach

- A. $+\infty$
- B. $-\infty$
- C. opposite infinities
- D. a finite hole

MCQ 2

No calculator

Which factor creates a hole at $x = 3$ when present in both numerator and denominator?

- A. $x + 3$
- B. $x - 3$
- C. $x^2 + 3$
- D. $3x - 1$

MCQ 3

No calculator

An uncanceled denominator factor $(x - a)^3$

- A. keeps the same sign on both sides
- B. changes sign across a
- C. creates a hole
- D. creates a zero

MCQ 4

No calculator

For $\frac{1}{x-5}$, $\lim_{x \rightarrow 5^-}$ equals

- A. $+\infty$
- B. $-\infty$
- C. 0
- D. 5

AP-style free response 1**No calculator - 6 points**

Let $R(x) = \frac{(x+4)(x-1)}{(x+2)^2(x-3)}$.

- A.** Identify zeros and vertical asymptotes.

- B.** Find both one-sided limits at -2 .

- C.** Find both one-sided limits at 3 .

- D.** Create a sign chart for all intervals.

AP-style free response 2

Calculator permitted - 6 points

For $t > 6$, a demand index is $D(t) = \frac{500}{t-6} + 80$.

A. Describe behavior as $t \rightarrow 6^+$.

B. Find and interpret long-run behavior.

C. Solve $D(t) = 180$.

D. Explain the contextual domain restriction and why $t = 6$ is not an output event.
